

COURSE SYLLABUS

CSC10108 – Data Visualization

1. GENERAL INFORMATION

Course name:	Data visualization
Course name (in Vietnamese):	Trực quan hóa dữ liệu
Course ID:	CSC10108
Knowledge block:	Required knowledge
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	
Prior-course:	
Instructors:	Dr. Bùi Tiến Lên

2. COURSE DESCRIPTION

The course provides basic knowledge about the visual representation. It enables you to transform data into persuasive and evidence-based visualizations. Visualizations are persuasive if they motivate actions in an intended audience. Visualizations are evidence-based if they are reproducible, functional, and truthful.

This course introduces and discusses the fundamental design principles and visualization technology that allows you to design, implement, and critique persuasive and evidence-based visualizations. In a data-rich environment, where decision-makers often drown in data but thirst for insight, mastering this course equips you with a moderate level of data literacy.

3. COURSE GOALS

At the end of the course, students are able to

Goal	Description	Program LOs
G1	Work with team members to analysis and visual data	
G2	Present English articles related to data visualization	

G3	Critical thinking	
G4	Apply data analysis in data visualization	
G5	Evaluate the meaning of a visualization	

4. COURSE OUTCOMES

CO	Description	I/T/U
G41	Understand concepts, principles, methods of data visualization	T
G4.2	Know how to map quantitative and qualitative items to visual items	T
G4.3	Use data analysis in data visualization	T
G4.4	Use visual representation for data types	T
G4.5	Use tools or software (JavaScript/D3, R, Python or Tableau)	T
G5.1	Know how to use data visualization in hypothesis tests	T
G1.1	Work in group	U
G2.1	Read and write English papers/articles	U
G3.1	Know how to evaluate a data visualization project	T

5. TEACHING PLAN

No	Topic	Course outcomes	Activities
1	Introduction to data visualization	2.1, 4.1	Lecture + discussion
2	Data and task abstraction	2.1, 4. 2	Lecture + discussion
3	Principle of visualization	2.1, 4.1	Lecture + discussion
4	The components in visualization	4.1	Lecture + discussion
5	Data analysis in visualization	4.1	Lecture + discussion
6	Multi-dimensional visualization	4.3	Lecture + discussion
7	Interactive visualization	4.1, 4.5	Lecture + discussion
8	Spatial visualization	4.4	Lecture + discussion
9	Evaluation	3.1, 5.1	Lecture + discussion
10	Other techniques		Lecture + discussion

6. ASSESSMENTS

ID	Name	Description	Ratio (%)
1	Lab		40%

2	Seminar	Start from week 6	10%
3	Midterm	In-class exam	20%
4	Final	Oral exam	30%
5	Activities		+10%

7. RESOURCES

Textbooks

1. Tamara Munzner, *Visualization Analysis and Design*, AK Peters Visualization Series, CRC Press, 2014.
2. Matthew O. Ward, Georges Grinstein, Daniel Keim, *Interactive Data Visualization: Foundations, Techniques, and Applications*, Second Edition, CRC Press, 2015.
3. Edward Tufte, *The Visual Display of Quantitative Information*, 2nd Edition, Graphics Press, 2001.
4. Edward Tufte, *Envisioning Information*, 3rd Edition, Graphics Press, 1990.
5. Colin Ware, *Information Visualization: Perception for Design*, 3rd Edition, Elsevier, 2013.

Others

N / A

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.
- **Students who are absent for mid-term exam or final exam and students who have 0% lab works scores are considered as unqualified for course completion.**